

```

a = 0.1;
A2 = 3;
x2 = A2 * exp(-a * n);

// Addition
x_add = x1 + x2;

// Multiplication
x_mul = x1 .* x2;

// Plot Sequence 1
subplot(4,1,1);
plot(n, x1);
xlabel("n (Discrete Time Index)");
ylabel("Amplitude");
title("Sequence 1: Sinusoidal Sequence");
xgrid();

// Plot Sequence 2
subplot(4,1,2);
plot(n, x2);
xlabel("n (Discrete Time Index)");
ylabel("Amplitude");
title("Sequence 2: Exponentially Decaying Sequence");
xgrid();

// Plot Addition
subplot(4,1,3);
plot(n, x_add);
xlabel("n (Discrete Time Index)");
ylabel("Amplitude");
title("Addition of Sequence 1 and Sequence 2");
xgrid();

// Plot Multiplication
subplot(4,1,4);
plot(n, x_mul);
xlabel("n (Discrete Time Index)");

```

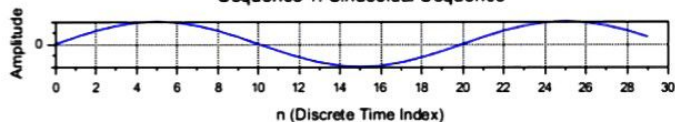
Graphic window number 0

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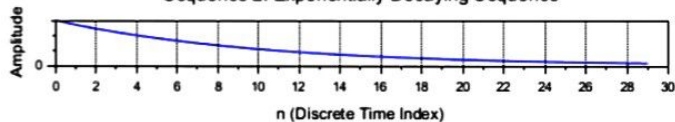


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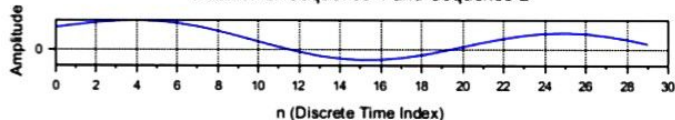
Sequence 1: Sinusoidal Sequence



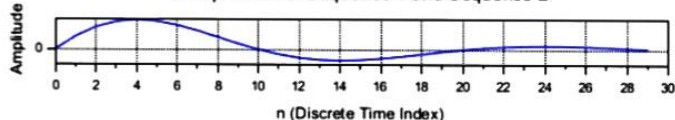
Sequence 2: Exponentially Decaying Sequence



Addition of Sequence 1 and Sequence 2



Multiplication of Sequence 1 and Sequence 2



```

1 // Test Case: Autocorrelation of a Rectangular Pulse Sequence
2 // Define parameters
3 N = 20; // Number of samples
4 x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5
5 // Compute the autocorrelation
6 R_xx = corral(x_n, x_n);
7 // Create a lag vector for plotting
8 lags = -N+1:N-1;
9 // Plot the input sequence
10 clf; // Clear the current figure
11 subplot(2,1,1); bar(0:N-1, x_n, 'b');
12 xlabel('n - (Discrete Time Index)');
13 ylabel('Amplitude');
14 title('Input Sequence: Rectangular Pulse Sequence');
15 // Plot the autocorrelation sequence
16 subplot(2,1,2); bar(lags, R_xx, 'r');
17 xlabel('Lag');
18 ylabel('Autocorrelation');
19 title('Autocorrelation of the Rectangular Pulse Sequence');
20

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